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Education

- 2022.08–**Doctor of Philosophy in Physics, Columbia University**
2027.05 (Expected) Dissertation: Search for Light Dark Matter near the ^8B Solar Neutrino Fog with XENONnT and a Krypton Background Measurement System at ppq Sensitivity
Advisor: Prof. Elena Aprile
- 2018.08–**Bachelor of Engineering (Major: Engineering Physics), Tsinghua University**
2022.08 Dissertation: Probing Coherent Elastic Neutrino–Nucleus Scattering from Reactor Neutrinos with LXeTPC and HPGe Detectors
Advisor: Prof. Fei Gao & Prof. Litao Yang

Selected Research Experience

XENONnT Dark Matter Project

- 2022–2026 **Solar ^8B CE ν NS and Light Dark Matter Search**
Led solar neutrino analyses and developed signal and background models for light dark matter searches in the neutrino fog
- Led event selection, background modeling, and four-dimensional likelihood inference for XENONnT's first indication of solar ^8B neutrino CE ν NS and subsequent 3.3σ evidence from three science runs
 - Corresponding author of two PRL papers (2024, 2026); main contributor to XENONnT's first light dark matter search in the neutrino fog (PRL, 2025)
- 2024–2026 **Analysis Coordinator of XENONnT**
Elected to coordinate collaboration-wide physics analyses, serving from October 2024 to February 2026
- Chaired analysis meetings, led analysis reviews, and coordinated working-group priorities and deadlines
 - Coordinated shared data provisioning and analysis infrastructure; brought detector, calibration, background, and analysis teams together to reach common scientific decisions
- 2025–now **Krypton Background Measurement System**
Developing XTRACT, a krypton-in-xenon measurement system targeting parts-per-quadrillion (ppq) sensitivity for rare-event background characterization
- Contribute to cryogenic design, construction, and commissioning of a cold-trap separation and residual-gas analysis system
 - Develop measurement automation and calibration to support krypton-background control for solar pp -neutrino and heavy WIMP searches
- 2024 **Convener of Computing and Analysis Tools Group**
Led the development of computing and analysis infrastructure supporting XENONnT physics analyses
- Developed and maintained HTC/HPC processing workflows involving 5 PB of I/O
 - Designed a more stable data-reprocessing workflow adopted for subsequent XENONnT analyses
- 2023–2024 **Toy-MC Inference Framework**
Developed alea, an open-source statistical inference framework for toy-Monte-Carlo confidence interval construction
- Designed modular inference workflows integrated with HPC computing for large-scale toy-Monte-Carlo studies
 - Adopted in XENONnT heavy WIMP, solar ^8B neutrino, and pp -neutrino analyses, as well as by external collaborations

2022–2023 **GPU-based LXe Response Simulation and Inference**

Developed appletree, an open-source GPU framework for liquid-xenon detector response simulation and Bayesian inference

- Ported the simulation from CUDA/C++ to JAX/Python, improving usability and maintainability
- Enabled MCMC-based detector response fitting subsequently used in XENONnT dark matter and neutrino analyses, including sub-keV light and charge yield calibration with a ^{88}YBe photoneutron source

Jinping Neutrino Experiment (JNE)

2019–2021 **PMT Waveform Analysis for Neutrino Detection**

Led a study of photon timing and charge reconstruction for neutrino and dark matter experiments, published in JINST (first author, 2022)

- Developed waveform reconstruction algorithms and a common framework to benchmark convolutional neural networks, deconvolution, and Bayesian methods
- Improved photon counting and timing reconstruction, including charge resolution in low-occupancy regimes

Gamma-Ray Integrated Detectors (GRID)

2018–2022 **CubeSat Gamma-ray Transient Events Analysis**

Developed a data processing pipeline for CubeSat gamma-ray burst (GRB) monitoring

- Built a pipeline subsequently used to detect over 40 GRBs, as of 2025
- Developed simulation tools for light curve analysis and localization algorithms

Selected Honors and Awards

2022.03 **Grand Prize of the 17th “Challenge Cup” Competition**, *Ministry of Education, etc.*

Student Extracurricular Academic and Technological Works Competition

Project: GRID-02 satellite payload calibration and scientific results

2021.12 & **Dongchang He Prize**, *Tsinghua University*

2019.12 Highest award for undergraduate of the Department of Engineering Physics, achieved twice

2020.12 **NAOC Fellowship**, *National Astronomy Observatory of China, Chinese Academy of Sciences*

Three winners per year in Tsinghua University, for students in the field of astronomy nationwide

2020.12 **BOEING Scholarship**, *Tsinghua University*

Excellent Comprehensive Scholarship of Tsinghua University

Talks

Conferences

2026.07 **Talk: Probing the Solar Boron-8 Neutrino Fog with XENONnT**, *Max-Planck-Institut für Kernphysik, Heidelberg, Germany*

Magnificent CEvNS 2026

2026.06 **Poster: Measurement of Solar ^8B Neutrino CEvNS with XENONnT**, *University of California, Irvine, Irvine, US*

Neutrino 2026: XXXII International Conference on Neutrino Physics and Astrophysics

2026.05 **Poster (People’s Choice Award): XTRACT: The Measurement of Krypton in Xenon with ppq Sensitivity (public PDF)**, *TRIUMF, Vancouver, Canada*

XeSAT 2026 - International Workshop on Applications of Noble Gas Xenon to Science and Technology

2025.06 **Talk: A measurement of the Solar Boron-8 Neutrino Fog with XENONnT**, *Centro Brasileiro de Pesquisas Físicas, Rio de Janeiro, Brazil (online)*

Magnificent CEvNS 2025

2025.06 **Talk: Data processing in XENONnT dark matter project**, *University of Wisconsin-Madison, Madison, US (online)*

Throughput Computing 2025

2024.10 **Talk: First Measurement of Solar ^8B Neutrinos via Coherent Elastic Neutrino-Nucleus Scattering with XENONnT**, *Blois, France*

Blois 2024: 35th Rencontres de Blois on “Particle Physics and Cosmology”

- 2024.07 **Talk: Search for Coherent Elastic Scattering of Solar ^8B Neutrinos in the XENONnT**, *Shandong University, Qingdao, China*
The 14th Conference of High Energy Physics Branch of Chinese Physical Society
- 2021.08 **Poster: Accurate and Robust PMT Waveform Analysis**, *Shandong University, Qingdao, China (online)*
The 13th Conference of High Energy Physics Branch of Chinese Physical Society
- 2021.06 **Poster: Optimized PMT Waveform Analysis**, *University of Minnesota Twin Cities, Minneapolis, US (online)*
The 28th International Workshop on Weak Interactions and Neutrinos (WIN2021)
- 2020.08 **Talk: PMT waveform analysis in neutrino experiments**, *Lawrence Berkeley National Laboratory, Berkeley, US (online)*
Dark-matter and Neutrino Computation Explored (DANCE) Machine Learning Workshop 2020

Seminars

- 2026.09 **Venturing Into the Neutrino Fog: Solar Boron-8 Neutrinos and Light Dark Matter with XENONnT**, *Massachusetts Institute of Technology, Cambridge, US*, Laboratory for Nuclear Science (LNS) Lunchtime Seminars at MIT
- 2026.09 **Turning a Dark Matter Detector into a Neutrino Observatory: Latest Results from XENONnT**, *Yale University, New Haven, US*, Nuclear Particle Astrophysics (NPA) Seminar at Yale
- 2026.08 **Probing the Solar ^8B Neutrino Fog with XENONnT**, *Lawrence Berkeley National Laboratory, Berkeley, US*, Institute for Nuclear and Particle Astrophysics (INPA) Seminar at LBNL
- 2026.07 **Probing the Solar ^8B Neutrino Fog with XENONnT**, *University College London, London, UK*, HEP seminar at UCL
- 2026.03 **Latest solar neutrino and light WIMP search with XENONnT**, *Washington University in St. Louis, St. Louis, US*, WUCAP
- 2025.12 **Latest solar neutrino and WIMP search results from the XENONnT experiment**, *Kavli Institute for the Physics and Mathematics of the Universe, Kashiwa, Japan*
- 2025.12 **Latest solar neutrino and WIMP search results from the XENONnT experiment**, *Institute for Basic Science, Daejeon, Korea*
- 2025.04 **Venturing into the Neutrino Fog with XENONnT**, *SLAC National Accelerator Laboratory, Menlo Park, US*
- 2024.09 **A measurement of solar neutrino with XENONnT experiment**, *Stony Brook University, Stony Brook, US*
- 2024.09 **First Measurement of Solar ^8B Neutrinos via Coherent Elastic Neutrino-Nucleus Scattering with XENONnT**, *Brookhaven National Laboratory, Upton, US*, Particle Physics Seminars at BNL

Schools

- 2026.08 **Quantum Probes for a Quantum Universe (54th SLAC Summer Institute)**, *SLAC National Accelerator Laboratory, Menlo Park, US*
- 2026.07 **Qiskit Global Summer School 2026: A Decade on the Cloud (Quantum Excellence Badge)**, *IBM Quantum (online)*
Completed hands-on quantum-circuit programming labs using Python and Qiskit

Mentoring

- 2025 **Khushi Vandra**, *Columbia University*, NSF Research Experiences for Undergraduates (REU)
S1 pulse shape discrimination in liquid xenon detectors

Teaching

2024 **Lecturer for XENONnT Computing Tutorial Series**, *Online*

Delivered biweekly tutorials on HTC/HPC workflows and analysis computing for XENONnT researchers

2022–2025 **Teaching Assistant**, *Columbia University*, Six Semesters

Intermediate Laboratory Work

Weekly lab sessions at E. K. A. advanced physics lab, covering topics in optics, quantum and particle physics

Internships

2024.07 & **Visiting Researcher**, *Laboratori Nazionali del Gran Sasso*, Assergi, Italy

2026.01 & Operation and calibration shifts for XENONnT

2026.07

2021 **Undergraduate Researcher**, *Columbia University*, Summer Undergraduate Research

Data analysis on the XENON detector, including several signal selection criteria

Supervisor: Prof. Elena Aprile

Outreach Activities

2021–2026 **Open-source Projects Maintainer**, *Columbia University*

Maintained open-source analysis tools, including appletree and alea, and supported their use across collaborations; appointed as a straxen maintainer in 2025

Professional Services

2024–now **Journal Referee**

Journal of Instrumentation, Physical Review Applied

References

Elena Aprile, *Centennial Professor of Physics*, Columbia University

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Laura Baudis, *Professor of Physics*, University of Zurich

XENON Collaborator, Email: laura.baudis@physik.uzh.ch

Luca Grandi, *Professor of Physics*, The University of Chicago

XENON Collaborator, Email: lgrandi@uchicago.edu

Kaixuan Ni, *Professor of Physics*, University of California San Diego

XENON Collaborator, Email: nikx@physics.ucsd.edu

Fei Gao, *Associate Professor of Physics*, Tsinghua University

B.Eng. advisor, Email: feigao@mail.tsinghua.edu.cn

For a complete publications list, please refer to my INSPIRE profile.

Selected Publications

- [1] E. Aprile et al. Probing the Solar ^8B Neutrino Fog with XENONnT. *Phys. Rev. Lett.*, 137(9):091807, 2026. **Corresponding Author**.

Main contribution: Led the combined analysis of three science runs and coordinated manuscript preparation, establishing 3.3σ evidence for solar ^8B neutrinos through coherent elastic neutrino–nucleus scattering and quantifying the impact of the neutrino fog on dark matter sensitivity.

- [2] E. Aprile et al. WIMP Dark Matter Search Using a 3.1 Tonne-Year Exposure of the XENONnT Experiment. *Phys. Rev. Lett.*, 135(22):221003, 2025. **Main Contributor**.

Main contribution: Developed the simulation and statistical inference frameworks underpinning XENONnT's WIMP search, including fast signal simulation and dedicated modeling of accidental-coincidence backgrounds.

- [3] E. Aprile et al. Low-energy nuclear recoil calibration of XENONnT with a ^{88}YBe photoneutron source. *Phys. Rev. D*, 113(11):112017, 2026. **Main Contributor.**
- Main contribution:* Developed the fast detector-response simulation and Bayesian inference framework for multiple-scatter nuclear recoils, enabling sub-keV light and charge yield calibration for solar-neutrino and light-dark-matter searches.
- [4] E. Aprile et al. First Search for Light Dark Matter in the Neutrino Fog with XENONnT. *Phys. Rev. Lett.*, 134(11):111802, 2025. **Main Contributor, Editors' Suggestion.**
- Main contribution:* Developed the signal and background models and four-dimensional likelihood framework that enabled XENONnT's first light dark matter search in the neutrino fog, including constraints on multiple dark matter interaction models.
- [5] E. Aprile et al. First Indication of Solar ^8B Neutrinos via Coherent Elastic Neutrino-Nucleus Scattering with XENONnT. *Phys. Rev. Lett.*, 133(19):191002, 2024. **Corresponding Author, Featured in Physics, Editors' Suggestion, APS Highlights of the Year.**
- Main contribution:* Led the analysis from low-threshold event selection and background modeling to four-dimensional likelihood inference, and coordinated manuscript preparation, establishing XENONnT's first indication of solar ^8B neutrinos.
- [6] C. Cai et al. Reactor neutrino liquid xenon coherent elastic scattering experiment. *Phys. Rev. D*, 110(7):072011, 2024. **Main Contributor.**
- Main contribution:* Developed the Geant4 detector simulation and sensitivity projections, informing detector and shielding design and quantifying sensitivity to the weak mixing angle and nonstandard neutrino interactions.
- [7] E. Aprile et al. First Dark Matter Search with Nuclear Recoils from the XENONnT Experiment. *Phys. Rev. Lett.*, 131(4):041003, 2023. **Main Contributor, Featured in Physics, Editors' Suggestion.**
- Main contribution:* Developed the data-driven accidental-coincidence background model and a boosted-decision-tree classifier with 65% background rejection at 95% signal acceptance for XENONnT's first nuclear-recoil dark matter search.
- [8] D. Xu et al. Towards the ultimate PMT waveform analysis for neutrino and dark matter experiments. *JINST*, 17(06):P06040, 2022. **First Author.**
- Main contribution:* Led algorithm development and established a common evaluation framework for PMT waveform reconstruction, systematically comparing convolutional neural networks, deconvolution, and Bayesian methods to improve photon counting and timing.